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Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Which of the following values of the log of odds [LOD score] is commonly used as a threshold level of significance in linkage studies?

Select one:

- More than 2
- More than 0.05
- More than 3
- Less than 0.05
- More than 1

Explanation:

A logarithm of the odds (LOD) score is calculated to determine the likelihood that two loci are linked at any particular genetic distance. The LOD score is calculated by dividing the likelihood of acquiring the data is the loci are linked at a given recombination fraction by the likelihood of acquiring the data if loci are unlinked. This step gives an odds ratio, and the log (base 10) of this odds ratio is the LOD score. The traditionally accepted LOD score threshold is 3.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 988.

The correct answer is: More than 3

Question **2**

Not answered

Marked out of 1.00

Flag question

Which of the following diagnoses is associated with a mutation in KISS1 gene?

Select one:

- Wernicke syndrome
- Cushing syndrome
- Williams syndrome
- Kallmann syndrome
- Addison syndrome

Explanation:

Kisspeptin and its receptor have emerged as the gatekeepers of puberty onset and the reproductive function. The KISS1 gene has been identified as involved in the X-linked or autosomal forms of Kallman syndrome (characterised by delayed or absent puberty and an impaired sense of smell).

Ref: Leka-Emiri S et al. The mystery of puberty initiation: genetics and epigenetics of idiopathic central precocious puberty (ICPP). Journal of Endocrinological Investigation. 2017 Aug 1;40(8):789-802.

The correct answer is: Kallmann syndrome

Question **3**

Not answered

Marked out of 1.00

Flag question

Which of the following chromosomal loci codes for Catechol-O-methyl transferase (COMT) enzyme?

Select one:

- chromosome 8q11
- chromosome 4q11
- chromosome 22q11
- chromosome 11q22
- chromosome Xq11

COMT gene resides in 22q11 locus.

The correct answer is: chromosome 22q11

Question **4**

Not answered

Marked out of 1.00

Flag question

Heritability is a measure of

Select one:

- Concordance of disease between twins
- Risk of disease in first degree relatives of probands
- Genetic contribution of disease in an individual
- Degree of penetrance of a disease
- Genetic contribution to phenotypic variation in population

The relative influence of genetic factors in determining the variance in a trait is expressed as heritability.

The correct answer is: Genetic contribution to phenotypic variation in population

Question **5**

Not answered

Marked out of

Choose the correctly paired techniques and genetic tools.

Select one:

1.00

Flag question

- Plasmid - Fluorescent in situ hybridisation
- Fluorescent probes - Polymerase chain reaction
- Heat denaturation - DNA cloning
- Oligonucleotide primers - Western blotting
- Vectors - DNA cloning

Explanation:

Once a DNA fragment is isolated (using restriction enzymes or PCR techniques), it is necessary to 'store' the fragment into a vector for future use and amplification. A DNA vector is a nucleic acid storage system that can hold an isolated DAN sequence of interest. Cloning vectors provide a backbone for the DNA insert to be reproduced and propagated in bacteria.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. pp. 234-235.

The correct answer is: Vectors - DNA cloning

Question **6**

Not answered

Marked out of 1.00

Flag question

Nucleosides consist of which of the following components?

Select one:

- Phosphorylated nucleotide
- Phosphate and nitrogenous base
- Pentose sugar, phosphate and nitrogenous base
- Pentose sugar and phosphate
- Pentose sugar and nitrogenous base

Genetic information is stored in the form of double-stranded DNA. DNA and RNA are the most important nucleic acids in the cellular machinery. These nucleic acids are composed of many nucleotides. Nucleotides are phosphorylated versions of nucleosides. Each nucleoside consists of two components: A nitrogenous base and a pentose sugar. Each strand of DNA is made up of a deoxyribose-phosphate backbone and a series of purine (adenine (A) and guanine (G)) and pyrimidine (thymine (T) and cytosine (C)) bases of the nucleic acid. The length of DNA is generally measured in numbers of base-pairs (bp).

The correct answer is: Pentose sugar and nitrogenous base

Question **7**

Not answered

Marked out of 1.00

Flag question

A 4-year-old child presents with developmental delay, growth retardation, generalised dystonia and an IQ of 60. On investigation, hyperuricaemia is noted. Which of the following is likely to be seen?

Select one:

- Insensitivity to pain
- Progressive dementia
- Self injurious behaviour

- Liver failure
- Hair loss

Repeated self-injurious behaviour is a common feature of Lesch-Nyhan syndrome in which hyperuricemia is also noted.

The correct answer is: Self injurious behaviour

Question **8**

Not answered

Marked out of 1.00

Flag question

Which of the following is true with regard to the genetic risk of schizophrenia when moving from a closer relative to a distant relative?

Select one:

- Risk initially increases, then falls
- Risk increases
- Risk depends on the sex of the relative
- Risk remains the same
- Risk decreases

For both schizophrenia and bipolar disorder there is a very rapid, non-linear decrease of risk when moving from a genetically identical individual (i.e. monozygotic co-twin where the risk is 50-60%), to an individual who shares half the genes (e.g. sibling, parent, dizygotic co-twin where risk is around 10%).(Retrieved from 'Phenotypic and genetic complexity of psychosis' http://bjp.rcpsych.org/content/190/3/200.full)

The correct answer is: Risk decreases

Question **9**

Not answered

Marked out of 1.00

Flag question

What is the risk of Alzheimer dementia in a patient heterozygous for ApoE4 compared to those with no ApoE4 allele?

Select one:

- Three times higher (OR=3)
- Nine times higher (OR=9)
- Same risk (OR=1)
- Twenty times higher (OR=20)
- Fifteen times higher (OR=15)

Explanation:

Certain allele combinations of the gene coding for ApoE, a cholesterol-carrying serum protein, confer substantial risk for Alzheimer disease (AD). Chromosome 19 encodes the gene for ApoE, which occurs in three forms: ApoE2, ApoE3, and ApoE4. Every person inherits one of these three alleles from each parent, giving each person an allele pair. Having two E4 alleles (being homozygous for E4) – and to a lesser degree, having one E4 allele (being heterozygous for E4) – significantly increases the risk of developing AD. Of individuals

carrying no E4 alleles, only about 20% develop AD; of those carrying one E4 allele, 50% develop AD (2-3x higher); and of those carrying two E4 alleles, 90% develop AD.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 122-123.

The correct answer is: Three times higher (OR=3)

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following genetic loci is associated with late onset senile dementia of Alzheimer's type?

Select one:

- Chromosome 1 Presenilin 2
- Chromosome 17 tau protein
- Chromosome 21 Beta amyloid
- Chromosome 14 Presenilin 1
- Chromosome 19 Apo E

Others are associated with presenile - early onset variety.

The correct answer is: Chromosome 19 Apo E

Question **11**

Not answered

Marked out of 1.00

Flag question

Which of the following is not true concerning Huntington's disease?

Select one:

- Age of onset is inversely related to disease severity
- Disease chromosome inherited from the father is unstable
- Trinucleotide repeat of 34-45 is seen in normal individuals
- Characteristic protein deposits form nuclear inclusions
- Maternal inheritance results in more severe and earlier onset disease

Spermatogenesis induces more instability in HD

The correct answer is: Maternal inheritance results in more severe and earlier onset disease

Question **12**

Not answered

Marked out of 1.00

Flag question

The best technique to detect a large sequence of DNA is

Select one:

- Electron microscopy
- PCR
- Southern blotting
- Linkage study

- Radiolabelling technique

Southern blotting was named after Edward M. Southern who developed this procedure at Edinburgh University in the 1970s. Southern blotting is designed to locate a particular sequence of DNA within a complex mixture. Fluorescent in situ hybridization (FISH) is a technique that exploits the principle of nucleic acid hybridization, in which single-stranded nucleic acids (usually DNA, sometimes RNA) interact with sufficiently similar, complementary sequences, to form hybrid complexes. This process helps to identify an unknown sequence and locate it on a given chromosome.

The correct answer is: Southern blotting

Question **13**

Not answered

Marked out of 1.00

Flag question

When the same disease phenotype is caused by mutations in different loci, this is called

Select one:

- Homogeneity
- Epistasis
- Allelic heterogeneity
- Locus heterogeneity
- Pleiotropy

Locus heterogeneity exists when the same disease phenotype can be caused by mutations in different loci. Locus heterogeneity becomes especially important when genetic testing is performed by testing for mutations at specific loci. For example, early-onset Alzheimer's could be caused by mutations in chromosome 1, 14 or 21.

The correct answer is: Locus heterogeneity

Question **14**

Not answered

Marked out of 1.00

Flag question

The total number of possible codons that can occur in human DNA is

Select one:

- 24
- 34
- 18
- 64
- 36

Explanation:

Amino acids are the building blocks of all proteins and are assembled as a long strand. Each sequence of three nucleotides of an mRNA molecule codes for one amino acid. The three nucleotides collectively form a ‘codon’. Because there are more possible codons (64) than amino acids, some codons translate into the same amino acids. The AUG codon translates into methionine and signals the start of translation. The UAA, UAG, and UGA codons do not

translate into amino acids, instead serving as a stop signal and terminating translation.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. pp. 214-215.

The correct answer is: 64

Question **15**

Not answered

Marked out of 1.00

Flag question

A 17-year-old boy's mother and paternal aunt both suffer from alcoholism. Compared to the general population, the risk of alcoholism in this boy is increased by a factor of

Select one:

- 8 times
- 3 times
- 2 times
- 16 times
- 4 times

Family studies show that the relative risk is increased by about 4-6 times for alcoholism if there is a strong family history. The majority of adoption studies show that the risk of alcoholism in adopted children is strongly correlated with their biological parents rather than adoptive parents (3-4 times higher).

The correct answer is: 4 times

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following syndromes is characterised by severe intellectual disability, rocker bottom feet, low-set ears, micrognathia and clenched hands?

Select one:

- Trisomy 18
- Velo-cardio facial syndrome
- Down syndrome
- Patau syndrome
- William syndrome

Explanation:

Edward syndrome, also called trisomy 18 syndrome, is an autosomal chromosomal disorder due to an extra copy of chromosome 18. It is one of the autosomal trisomy syndromes, secondary in frequency only to trisomy 21. Edward syndrome is characterised by severe intellectual disability and rocker bottom feet, low-set ears, micrognathia (small jaw), congenital heart disease, clenched hands, and a prominent occiput. It occurs at a frequency of 1:8000 and often death occurs within 1 year of birth.

The correct answer is: Trisomy 18

Question **17**

Not answered

Marked out of 1.00

Flag question

Which of the following phenomena explain most of the epigenetic variations?

Select one:

- Mutations and polymorphism
- Aneuploidy
- DNA methylation and histone modification
- Polymorphism and microdeletions
- Splicing errors and methylation

DNA methylation and histone modification explain most the of epigenetic variations discovered to date. Crow has argued for long that epigenetic defects explain most of the concordance seen in schizophrenia; according to Crow, the hemispheric laterality and language specialisation unique to human brains is the source of schizophrenic defect and it can be ascertained only through an epigenetic enquiry. Qiu J (2006) Epigenetics: unfinished symphony. Nature, 441, 143-145.

The correct answer is: DNA methylation and histone modification

Question **18**

Not answered

Marked out of 1.00

Flag question

When compared with the general population, the range of full-scale IQ in people with Turner syndrome is shifted down by

Select one:

- 30 points
- 20 points
- 0 points
- 40 points
- 10 points

Explanation:

Turner syndrome is a sex chromosome monosomy (karyotype 45, XO) with female phenotype. Individuals with Turner syndrome generally have a normal IQ, and intellectual disability is rare. There may be specific deficits of visuospatial learning.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 817.

The correct answer is: 0 points

Question **19**

Not answered

Marked out of 1.00

Flag question

Which of the following types of Down syndrome could be inherited with equivalent likelihood from either of the parents?

Select one:

- Monosomy 21
- Unbalanced translocation

- Trisomy 21
- Mosaicism
- Robertsonian translocation

Explanation:

Translocation accounts for trisomy 21 in approximately 5% of affected individuals, usually t(14;21) or t(21;21). In translocation, a fusion occurs, of two chromosomes, resulting in a total of 46 chromosomes, despite the presence of an extra chromosome 21. This is usually inherited, and the translocated chromosome may be found in unaffected parents and siblings. Asymptomatic carriers have only 45 chromosomes.

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Feb 6;6(1):1-20.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 101.

The correct answer is: Robertsonian translocation

Question **20**

Not answered

Marked out of 1.00

Flag question

Which of the following can be used to study the abnormalities in the candidate genes associated with schizophrenia in patients with psychosis?

Select one:

- Chromosomal count
- Amount of methylation
- Single nucleotide polymorphisms
- Heterochromatin material
- Translocation

Genome-wide association studies are based on linkage disequilibrium (LD), using several hundred thousand marker single nucleotide polymorphisms or SNPs.

The correct answer is: Single nucleotide polymorphisms

Question **21**

Not answered

Marked out of 1.00

Flag question

In a closed islandic population, frequency of genotype HH is 50, hh is 10 while Hh is 40. What is the gene frequency of H?

Select one:

- 14
- 0.7
- 70
- 7
- 1.4

Out of 200 gene copies in 100 people, H occurs 50+50+40 = 140 times. Hence gene frequency of H is 140/200 = 0.7

The correct answer is: 0.7

Question **22**

Not answered

Marked out of 1.00

Flag question

You are seeing a 55-year-old man in the outpatients' clinic. He has a history of gradual progressive memory loss, and his MMSE is 20 out of 30. The gene implicated for his condition codes for

Select one:

- Glutamate
- Presenilin-1
- Neuronal Sortilin related receptors (SORL1)
- Dysbindin
- Apolipoprotein - E4

Explanation:

Mutations in the presenilin-1 gene located on chromosome 14 have been implicated in young-onset Alzheimer disease, as in this case.

Risk and protective factors	Impact on risk of dementia and Alzheimer disease (AD)
Age and gender	Largest risk factor for dementia and AD is advancing age Women have higher prevalence and incidence of dementia than men
Education and intellectual stimulation	Lower educational levels associated with higher incidence of dementia and AD Cognitive reserve hypothesis: education and intellectual stimulation extend neural reserve; brain can withstand more insults before developing dementia
Psychosocial factors	Increased social engagement associated with lower rates of cognitive decline Depression is possible risk factor for developing dementia
Genetic susceptibility	First-degree relatives of people with AD have higher risk of developing AD Young-onset AD: main genetic cause is mutations in three genes - amyloid precursor protein (APP), presenilin 1, and presenilin 2 Late-onset AD: complex interplay between genetic predisposition and environmental risk factors; apolipoprotein E (APOE) E4 allele confers susceptibility while APOE E2 allele is protective
Vascular risk factors	Smoking, high cholesterol, high blood pressure, and diabetes mellitus are all associated with increased risk of AD
Hearing loss	Increased risk of dementia in people with hearing loss

Adapted from Lilford and Hughes 2020

SPMM COURSE

Ref: Lilford P, Hughes JC. Epidemiology and mental illness in old age. BJPsych Advances. 2020 Mar;26(2):92-103.

The correct answer is: Presenilin-1

Question **23**

Not answered

Marked out of 1.00

Which of the following phenomena takes place during interphase?

Select one:

- Preparation for cell division

Flag question

- Cell division
- Quiescence
- Cell union
- Nuclear division

Cell division is a process by which cells reproduce " here, a sequence of steps enable the replicated genetic material in a parent cell to be equally distributed to two daughter cells. Before a dividing cell enters mitosis, it undergoes a period of growth called interphase. Interphase is the 'holding' stage between two successive cell divisions. In this stage, the cell replicates its genetic material and organelles in preparation for division. It is the longest phase, and all steps in cell cycle except stage (M) together constitute interphase.

The correct answer is: Preparation for cell division

Question **24**

Not answered

Marked out of 1.00

Flag question

Which of the following condition results from a genetic abnormality in the locus coding progranulin?

Select one:

- Primary progressive aphasia
- Alzheimer disease
- Tay Sach disease
- Prion disease
- Parkinson disease

Explanation:

The spectrum of GRN frontotemporal dementia (GRN-FTD) includes the behavioural variant (bvFTD), primary progressive aphasia (further categorised as progressive non-fluent aphasia and semantic dementia), and movement disorders with extrapyramidal features such as parkinsonism and corticobasal syndrome. GRN encodes progranulin, a glycoprotein. Progranulin mRNA is widely expressed in rapidly cycling epithelial cells, in the immune system, and in neurons such as cerebellar Purkinje cells, suggesting an important function in these tissues. Progranulin is involved in multiple physiologic processes such as cellular proliferation and survival as well as tissue repair, and pathologic processes including tumorigenesis.

Ref: Hsiung GYR et al. 2007 Sep 7 [updated 2020 Feb 6]. In: Adam MP, Ardinger HH, Pagon RA, Wallace SE, Bean LJH, Stephens K, Amemiya A, editors. GeneReviews® [Internet]. Seattle (WA): University of Washington, Seattle; 1993–2020. PMID: 20301545.

The correct answer is: Primary progressive aphasia

Question **25**

Not answered

Marked out of 1.00

Which of the following terms describes deletion or insertion mutations that take place in multiples of triplet codons?

Select one:

- Inframe mutations

Flag question

- Substitutions
- Frame shift mutations
- Transitions
- Inversions

According to the effect on triplet sequence, mutations could be frame shift or in-frame. In frame shift mutations, the deletion or insertion is not in multiples of three codons e.g. a segment of 5 bases deletion mutations. This leads to a shift in triplet reading frame with variable results. Inframe mutation refers to changes happening in multiples of 3 bases, with no disturbances in the actual reading frame.

The correct answer is: Inframe mutations

Question **26**

Not answered

Marked out of 1.00

Flag question

The process by which one gene gives rise to more than one protein is

Select one:

- Inverse mutation
- Transcription
- Alternate splicing
- Epistasis
- Translocation

In human DNA, the genetic information that includes the code for making a protein is located in fragments (exons, red boxes), which are interrupted by non-coding fragments (introns, green boxes). By the process of alternative splicing, the introns are removed, and the exons spliced together in different combinations, generating different messenger RNAs (mRNA) that are decoded (translated) into distinct proteins. In humans, it is estimated that alternative splicing occurs in more than 60% of genes, thus abolishing the one gene, one protein dogma long held in medical genetics.

The correct answer is: Alternate splicing

Question **27**

Not answered

Marked out of 1.00

Flag question

Which of the following statements about zygosity in twins is true?

Select one:

- Dizygotes share all except few genes
- Monozygotes are formed from two sperms fertilizing one ovum
- Dizygotes and monozygotes have same amount of environmental sharing
- Monozygotes share more uterine environment than dizygotes
- Anatomical similiarity is sufficient to establish zygosity

Monozygotic (MZ, or "identical") twins are formed when an embryo is cleaved during early development. The result is two genetically identical embryos. Dizygotic (DZ, or "fraternal") twins are the result of the fertilization of two different ova by two different sperm cells. DZ twins are genetically the same as non-twin siblings, sharing only 50% of their genes. Approximately 70% of monozygotic twins are monochorionic while dizygotes almost always have separate chorions/amnions. Hence, monozygotes share more uterine environment than dizygotes.

The correct answer is: Monozygotes share more uterine environment than dizygotes

Question **28**

Not answered

Marked out of 1.00

Flag question

A young girl with learning difficulties is diagnosed to have trisomy X. Her chromosomes were karyotyped to diagnose the genetic defect. How many Barr bodies would she have in her somatic cells?

Select one:

- Information insufficient to answer the question
- One
- Two
- None
- Three

X inactivation occurs very early in the development of female embryos. When an X chromosome is inactivated, it could be visualized under the microscope as a highly condensed Barr body in the nuclei of interphase cells. All X chromosomes in a cell are inactivated except one, irrespective of original number of X chromosomes in a cell. Thus females with trisomy X will have two Barr bodies. X inactivation occurs via DNA methylation.

The correct answer is: Two

Question **29**

Not answered

Marked out of 1.00

Flag question

Mrs. Brown gives birth to twins in her first delivery. The twins look alike with respect to their hair colour but look different with respect to the colour of iris. Which of the following statement is most likely with the available information?

Select one:

- They are dizygotic twins
- They are monozygotic twins
- They are concordant for iris colour
- They are identical twins
- They are concordant for hair colour

Two individuals are concordant if they share the same trait; if they do not share the trait, they are discordant. For a trait that is completely genetic, one would expect a concordance of 100% for MZ twins because they share all of their genes. The concordance rates for DZ twins should be only about 50%. In contrast, a trait that has no genetic basis should have equivalent concordance rates for MZ and DZ twins.

The correct answer is: They are concordant for hair colour

Question **30**

Not answered

Marked out of 1.00

Flag question

Which of the following blotting techniques can be used to visualize restriction fragment length polymorphisms?

Select one:

- Southern blot
- Northern blot
- Dot blot
- Eastern blot
- Western blot

Explanation:

A Southern blot is used to identify if a known DNA sequence is present in a sample by using a specific probe to identify the sequence in a large mixture of other DNA fragments. A DNA sample is digested with restriction enzymes, and the mixture of fragments is separated by gel electrophoresis, then transferred to a membrane. The membrane is incubated with radioactive probes capable of hybridising to specific nucleic acid sequences. The bands can then be visualised using autoradiography.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. pp. 239-240.

The correct answer is: Southern blot

Question **31**

Not answered

Marked out of 1.00

Flag question

Variation of a genetic code across the population without any associated overt phenotypic change is best described as

Select one:

- Mutation
- Epigenesis
- Polymorphism
- Gene silencing
- Point deletion

Polymorphisms are natural variations in a gene, DNA sequence, or chromosome that have no adverse effects on the individual and occur with fairly high frequency in the general population. Gene silencing refers to the epigenetic processes of gene regulation whereby the "switching off" of a gene takes place by a mechanism other than genetic modification. This occurs when RNA is not translated to a protein downstream (Retrieved from Genetics Home Reference, http://ghr.nlm.nih.gov/glossary=polymorphism (accessed April 13, 2015).).

The correct answer is: Polymorphism

Question **32**

Not answered

Marked out of 1.00

Flag question

Which of the following methods can be used to detect the presence of abnormal trinucleotide expansions?

Select one:

- Northern blotting
- Karyotyping
- Southern blotting
- PCR
- FISH

Karyotyping results in a picture of someone's chromosomes, taken from a single cell. A karyotype can reveal large segments of missing or extra genetic material (e.g., in Down syndrome, there are three copies of chromosome 21; a female with Turner syndrome only has one X chromosome instead of two). FISH uses a protein, called a probe, to "stick" to a known sequence of DNA (usually a known mutation). If that sequence is present in a patient's sample, the probe will bind to it and light up under a fluorescent microscope. FISH also can be used to detect chromosome rearrangements, marker chromosomes (extra pieces of unidentified chromosomal material), and duplications or deletions of large pieces of DNA (Excerpt retrieved from http://www.nchpeg.org/). PCR is commonly used in medical and biological research labs to copy segments of DNA (called DNA amplification) so that sufficient quantities are available for further testing (e.g., forensic, medical, research). Relatively simple PCR-based assays can be used to determine the presence of the trinucleotide expansions. The size of the PCR product can be evaluated by polyacrylamide gel electrophoresis. The product length is detected by use of a radiolabeled primer to generate a radiolabeled product, e.g. a radiolabeled (CAG)n or (CTG)n oligonucleotide (Retrieved from http://www.clinchem.org/content/49/10/1726.long).

The correct answer is: PCR

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following nuclear components undergo removal during splicing?

Select one:

- Histones
- Introns
- Trinucleotide repeats
- Exons
- Transfer-RNA

An intron is any nucleotide sequence within a gene that is removed by RNA splicing to generate the final mature RNA product of a gene

The correct answer is: Introns

Question **34**

Not answered

The investigation of choice used in the diagnosis of Klinefelter's syndrome is

Marked out of 1.00

Flag question

Select one:

- Fetal topography
- Fetal Ultrasound
- Amniocentesis
- Karyotyping
- Chorionic villus sampling

A karyotyping is frequently used to confirm the diagnosis of Klinefelter's syndrome. In this procedure, white blood cells from the infant are separated from the plasma, mixed with a tissue culture medium, incubated, and checked for chromosomal abnormalities, such as the presence of an extra X chromosome. Diagnosis can also be made prenatally via chorionic villus sampling or amniocentesis tests in which fetal tissue is extracted, and the fetal DNA is examined for genetic abnormalities.

The correct answer is: Karyotyping

Question **35**

Not answered

Marked out of 1.00

Flag question

Which of the following conditions is an example of aneuploidy?

Select one:

- Thalassaemia
- Huntington disease
- Absence seizures
- Phenylketonuria
- Down syndrome

Explanation:

Trisomy 21, the presence of a supernumerary chromosome 21, results in a collection of clinical features commonly known as Down syndrome. Down syndrome is among the most genetically complex of the conditions that are compatible with human survival post-term, and the most frequently survivable autosomal aneuploidy (presence of an abnormal number of chromosomes).

Ref: Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Feb 6;6(1):1-20.

The correct answer is: Down syndrome

Question **36**

Not answered

Marked out of 1.00

Flag question

Which of the following refers to the production of new copies of RNA from DNA?

Select one:

- Translation
- Synapsis
- Replication
- Transcription

- Modification

Explanation:

Transcription is the synthesis of a messenger RNA (mRNA) molecule from a DNA template. Translation is the process of using an mRNA molecule as a template to produce a protein.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. pp. 212-214.

The correct answer is: Transcription

Question **37**

Not answered

Marked out of 1.00

Flag question

An autosomal recessive disorder shows Hardy-Weinberg equilibrium. It has an incidence of 1 in 4900. The frequency of carriers of the disease is

Select one:

- 1 in 30
- 1 in 70
- 1 in 35
- 1 in 10
- 1 in 40

According to Hardy-Weinberg principle, the carrier frequency is twice the square root of the frequency of a disease.

The correct answer is: 1 in 35

Question **38**

Not answered

Marked out of 1.00

Flag question

The function of the enzyme RNA polymerase is best described as

Select one:

- Synthesis of multiple copies of damaged RNA
- Synthesis of DNA from RNA template
- Synthesis of DNA from DNA template
- Synthesis of RNA from DNA template
- Synthesis of RNA from RNA template

RNA Polymerase is an enzyme that catalyzes the synthesis of RNA from ribonucleoside triphosphates in the presence of a DNA template.

The correct answer is: Synthesis of RNA from DNA template

Question **39**

Not answered

Following synthesis, proteins inside the cell are stored temporarily in

Marked out of 1.00

Flag question

Select one:

- Ribosomes
- Nucleus
- Nucleolus
- Golgi complex
- Cilia

Explanation:

The golgi complex (or golgi apparatus) is a complex of stacks of flattened membrane sacs (golgi vesicles) in the cytoplasm of most cells, involved in the transport of lipids and proteins.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Golgi complex

Question **40**

Not answered

Marked out of 1.00

Flag question

In which of the following patterns of inheritance does skipping one generation but reappearing on the next is commonly seen?

Select one:

- Translocations
- Mitochondrial inheritance
- Autosomal dominant
- X linked dominant
- Autosomal recessive

Autosomal recessive disorders manifest themselves only when an individual is homozygous for the disease allele, i.e. both chromosomes carry the mutated gene. In this case, the parents are generally unaffected, healthy but carriers (heterozygous for the disease allele). There is usually no family history, although the defective gene is passed from generation to generation (skipping). The offspring of an affected person will be healthy heterozygotes unless the other parent is also a carrier. If carriers marry, the offspring have a 1 in 4 chance of being homozygous and affected and a 1 in 2 chance of being a carrier, and a 1 in 4 chance of being genetically normal. Consanguinity increases the risk. Often the abnormal gene in autosomal recessive diseases codes for enzymatic proteins.

The correct answer is: Autosomal recessive

Question **41**

Not answered

Marked out of 1.00

Flag question

Which of the following is an autosomal dominant disorder?

Select one:

- Friedreich ataxia
- Phenylketonuria
- Noonan syndrome
- Hurler syndrome

- Tay-Sachs disease

Explanation:

Noonan syndrome is **autosomal dominant**, occurring in 1:1000-1:2000. Males and femalesare equally affected. Clinical features include varying degrees of intellectual disabillity (from none to severe), short stature (80%), cardiac abnormalities (>80%), hepatosplenomegaly (25%), and distinctive facies. While a number of genes (including PTPN11, SOS1, and KRAS) have been identified that can cause Noonan syndrome, it remains a clinical diagnosis.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 810.

The correct answer is: Noonan syndrome

Question **42**

Not answered

Marked out of 1.00

Flag question

What is the likelihood (in percentage) of a man acquiring Huntington's disease if his father has the illness and carries the mutant chromosome?

Select one:

- 33
- 75
- 50
- 25
- 100

Huntington's disease is transmitted by an autosomal dominant inheritance pattern. If one parent is an affected carrier, the likelihood of transmission to any given child is 50%. The protein huntingtin is coded on the short arm of chromosome 4.

The correct answer is: 50

Question **43**

Not answered

Marked out of 1.00

Flag question

Which of the following pairing among nucleotides does not occur in human DNA sequences?

Select one:

- AT pair
- GC pair
- TG pair
- CG pair
- TA pair

Explanation:

A DNA molecule is a long polymer made up of repeating units called nucleotides. These nucleotides are composed of a phosphate group, a sugar molecule, and a nitrogenous base.

The phosphate and sugar molecules form the backbone of a DNA polymer. The two strands of DNA are stabilised by hydrogen bonds between the nitrogenous bases attached to the two strands. There are four types of nitrogenous bases found in DNA: adenine (A), thymine (T), guanine (G), and cytosine (C). A base on one strand of DNA forms a bond with only one type of base on the opposite strand: A only forms bonds with T, and C only forms bonds with G. (A TG pairing is not possible.)

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 211.

The correct answer is: TG pair

Question **44**

Not answered

Marked out of 1.00

Flag question

The number of daughter cells at the end of a complete meiosis of one human cell is

Select one:

- 2
- 5
- 4
- 3
- 1

Explanation:

Meiosis is a form of cell division in which the nucleus divides into four daughter nuclei, each containing half the chromosomes of the parent nucleus. It consists of two consecutive divisions: during the first, homologous chromosomes pair up, exchange genetic material by crossing over ('recombination'), then during the anaphase part of the cycle move apart by disjunction to form separate haploid nuclei. During the second phase, these daughter nuclei divide by mitosis (with a further anaphase), producing four haploid cells.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: 4

Question **45**

Not answered

Marked out of 1.00

Flag question

Which of the following describes the process by which introns are removed from freshly made RNA sequences?

Select one:

- Post translational modification
- Exocytosis
- Cytokinesis
- Splicing
- Synapsis

Transcripted RNA initially contains the 'junk' sequences - introns that do not code for polypeptides. This unprepared RNA is called heterogeneous nuclear RNA or hnRNA. This hnRNA then undergoes splicing aided by nucleosomes in the nucleus to remove non-coding

sequences and results in messenger RNAs (mRNA).

The correct answer is: Splicing

Question **46**

Not answered

Marked out of 1.00

Flag question

Neuregulin-1 was identified as a candidate gene for schizophrenia after typing 950 microsatellite markers in 110 Icelandic patients. Which of the following approach was used to complete the above?

Select one:

- Family studies
- Sib pair analysis
- Whole genome scan
- Association studies
- Twin studies

Stefansson et al. found that neuregulin-1 is a candidate gene for schizophrenia after typing 950 microsatellite markers covering the whole genome in 110 Icelandic patients for whom they had reconstructed their genealogical relationships. This was done using whole genome scan, which is a type of linkage analysis, where markers placed at regular intervals covering the whole genome, are typed. It is tedious but often the first approach when no genetic information is available about a particular phenotype (Advanced glossary on genetic epidemiology. Retrieved from http://jech.bmj.com/content/57/8/562.full.pdf).

The correct answer is: Whole genome scan

Question **47**

Not answered

Marked out of 1.00

Flag question

Mr X is a tall man with low IQ and behaviour problems. Identify the type of sex chromosome disorder?

Select one:

- Down syndrome
- XYY male
- Klinefelter syndrome
- Turner syndrome
- Trisomy X

XYY males have a karyotype of 47, XYY. This condition is seen in 1:1000 male births. Behavioural problems are commonly seen. IQ may be slightly lower than average. Studies on the population of men in maximum-security psychiatric hospitals in the 1960s seemed to show an excess of men with this karyotype as well as men with Klinefelter's syndrome.

The correct answer is: XYY male

Question **48**

Which of the following is responsible for dementia seen in adults with Down's syndrome?

Not answered

Marked out of 1.00

Flag question

Select one:

- Loss of genetic material corresponding to presenilin -1
- Loss of genetic material in chromosome 21
- Genetic material lost from chromosome 14
- Loss of genetic material corresponding to Apoe4
- Extra genetic material in chromosome 21

Chromosome 21 is implicated in both Down's and Alzheimer's. The extra genetic material in trisomy causes a pathology similar to dementia.

The correct answer is: Extra genetic material in chromosome 21

Question **49**

Not answered

Marked out of 1.00

Flag question

Which of the following steps is NOT a part of polymerase chain reaction (PCR)?

Select one:

- Extending primers in opposite directions
- Vector based replication of DNA
- Denaturing DNA by heat
- Annealing DNA by cooling
- Amplifying copies of DNA

The PCR technique has three steps. A. Double-stranded genomic DNA is denatured by heat into single-stranded DNA. The reaction is then cooled to favour DNA annealing, and the primers bind to their target DNA. B. DNA polymerase is used to extend the primers in opposite directions using the target DNA as a template. After one cycle, there are two copies of double-stranded DNA, after two cycles there are four copies, and this number rises exponentially with the number of cycles. C. The cycling is set to produce necessary number of amplifications. Note that the vector-based replication is used for DNA cloning, not PCR.

The correct answer is: Vector based replication of DNA

Question **50**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a step in polymerase chain reactions?

Select one:

- Cycling
- Annealing the DNA
- Extending the primers
- Gel electrophoresis
- Denaturing the DNA

Minute amounts of DNA can be amplified over a million times using an in vitro technique called polymerase chain reaction. The technique has three steps. Double-stranded genomic DNA is denatured by heat into single-stranded DNA. The reaction is then cooled to favour DNA annealing, and the primers bind to their target DNA. DNA polymerase is used to extend the primers in opposite directions using the target DNA as a template. After one cycle, there are two copies of double-stranded DNA, after two cycles there are four copies, and this number rises exponentially with the number of cycles. The cycling is set to produce necessary number of amplifications. Gel electrophoresis is not an inherent part of PCR.

The correct answer is: Gel electrophoresis

Question **51**

Not answered

Marked out of 1.00

Flag question

A patient is referred with a diagnosis of schizophrenia. Chromosome analysis shows a 22q11 deletion confirming a diagnosis of Di George syndrome. Which of the following is NOT compatible with this diagnosis?

Select one:

- pharyngeal insufficiency
- supra-valvular aortic stenosis
- hypercalcaemia
- fish mouth
- severe learning difficulties

In velo-cardio-facial or DiGeorge syndrome, hypoparathyroidism leading to hypocalcemia (observed in 60% of patients) usually begins in the neonatal period, occasionally manifesting in the form of tetany or tonic convulsions.

The correct answer is: hypercalcaemia

Question **52**

Not answered

Marked out of 1.00

Flag question

Molecular genetic findings in psychiatric disorders are consistent with which of the following models?

Select one:

- Single genes of large effect
- Multiple genes of small effect
- Single genes of small effect
- Multiple genes of large effect
- None of the listed options

Multiple genes - small effect model proposes that diseases such as schizophrenia result from the co-action of multiple (ranging in principle from a few to many thousand) common variants (`polymorphisms'), each of which has a small effect on illness susceptibility. When an individual inherits several, or many, susceptibility variants together, they have a sizable influence on disease risk. Hence the mutations or polymorphisms are not sufficient by themselves to cause disease, but they occur very commonly so they can interact in combinations and produce the disease. This model is more popular currently and forms the

basis of association and linkage studies being carried out widely.

The correct answer is: Multiple genes of small effect

Question **53**

Not answered

Marked out of 1.00

Flag question

An aborted foetus was found to have 69 chromosomes in each cell. Which of the following terms best describes the abnormality?

Select one:

- Diploid
- Euploid
- Haploid
- Monoploid
- Triploid

When cells possess chromosomal numbers different from normal diploidy, they are called aneuploid cells. Aneuploidy can occur in single numbers e.g. trisomy 21, trisomy 18, monosomy of Turner's, etc. Very rarely, the entire chromosome set will be present in more than two copies, so the individual may be triploid rather than diploid and have a chromosome number of 69. Triploidy and tetraploidy (four sets) result in spontaneous abortion.

The correct answer is: Triploid

Question **54**

Not answered

Marked out of 1.00

Flag question

The risk of unipolar depression in a first degree family member with bipolar disorder is given by

Select one:

- 6-7 times higher
- 2-3 times higher
- 4-5 times higher
- 8-9 times higher
- 10 times higher

Lifetime risk rates of unipolar disorder in a member of the general population unrelated to anyone with affective disorders is 5-10%. In relatives of a bipolar proband, the risk, of the unipolar disorder is A. first-degree relative 10-20% (relative risk = 2-3times); B. monozygotic co-twin 15-25% (relative risk = 3-5 times). Also, note that the relatives of bipolar patients have a risk of 20-30% for 'any affective disorder' in their lifetime. Note that the risk to offspring with both parents affected by bipolar disorder is 50-65% risk for bipolar and 50-75% risk for any affective disorder. (From http://www.nchpeg.org)

The correct answer is: 2-3 times higher

Question **55**

Which of the following enzymes mediate the binding of tRNA with amino acids?

Not answered

Marked out of 1.00

Flag question

Select one:

- RNA polymerase
- Protein synthetase
- Amino acid dehydrogenase
- Aminoacyl synthetase
- Reverse transcriptase

While syntheszing protein chains, the correct amino acid is added to the coding *tRNA* by a specific enzyme called aminoacyl-tRNA synthetase. This synthetic process is called aminoacylation.

The correct answer is: Aminoacyl synthetase

Question **56**

Not answered

Marked out of 1.00

Flag question

A distraught mother whose first child is diagnosed recently with autistic disorder wants to know the relative risk of the same disorder to recur in her next child. The correct answer is

Select one:

- 3 times
- 50 times
- 10 times
- 4 times
- 100 times

Autism is thought to be strongly genetic.

The correct answer is: 50 times

Question **57**

Not answered

Marked out of 1.00

Flag question

Heritability of bipolar disorder is

Select one:

- 75-85%
- 55-65%
- 65-75%
- 85-100%
- 45-55%

The overall heritability of the bipolar disorder varies between 0.75 to 0.80.

The correct answer is: 75-85%

Question **58**

Not answered

Marked out of 1.00

Flag question

Which of the following correctly describes gene mapping?

Select one:

- A strategy by which parental identity is established in case of disputes
- A method by which genes are distributed during mitosis
- A method by which genes are distributed in meiosis
- A strategy that permits finding the chromosomal location of genes
- A process by which ethnographic genetic distribution is studied

Gene mapping refers to any strategy that permits finding the chromosomal location of one or more genes, often related to a disease. Genetic mapping of disease genes is a very useful method because it does not require any knowledge of a gene's function to find the chromosomal location initially. Once located then the identity of the disease gene could be dissected. Not all genetic studies are aimed at gene mapping; certain simpler designs are primarily aimed at demonstrating the presence or absence of a genetic influence in the aetiology of a disease or trait. These include family studies, twin studies, and adoption studies. Gene mapping studies involve linkage analysis, sib-pair analysis and to some extent allelic association studies.

The correct answer is: A strategy that permits finding the chromosomal location of genes

Question **59**

Not answered

Marked out of 1.00

Flag question

If both parents suffer from an autosomal recessive disorder, the probability of each child being affected is

Select one:

- 1:1
- 1:2
- 1:4
- 1:3
- 1:5

Autosomal recessive disorders manifest themselves only when an individual is homozygous for the disease allele; i.e. both chromosomes carry the mutated gene. So if both parents suffer from a disorder, the probability of each child being affected is 100% or 1:1. If carriers marry, the offspring has a 1 in 4 chance of being homozygous and affected and a 1 in 2 chance of being a carrier, and a 1 in 4 chance of being genetically normal. Consanguinity increases the risk.

The correct answer is: 1:1

Question **60**

Not answered

Marked out of 1.00

Choose the correct statement concerning Prader-Willi syndrome

Select one:

- Associated with short stature

Flag question

- Occurs only in males
- Social anxiety is a common feature
- Affected children have a normal IQ
- Average life expectancy of 6 to 8 years

Patients with PWS are often short statured and obese with learning disabilities.

The correct answer is: Associated with short stature

Question **61**

Not answered

Marked out of 1.00

Flag question

Which one of the following processes can inactivate a gene?

Select one:

- Unwinding of DNA strands
- Uncoiling of chromosome
- Methylation
- Condensation
- Crossing over

DNA methylation is a source of epigenetic modifications.

The correct answer is: Methylation

Question **62**

Not answered

Marked out of 1.00

Flag question

Which of the following aminoacids constitute the initiation sequence of protein translation?

Select one:

- Tyrosine
- Isoleucine
- Threonine
- Valine
- Methionine

Translation includes three steps - initiation, elongation and termination. The ribosome contains two sites: Peptidyl P site where methionine containing tRNA initially binds and aminoacyl A site where each new incoming tRNAs with activated amino acids can bind. In elongation step, amino acids are added one by one in a sting like fashion to produce proteins. Chain termination is signaled by one of the three codons UAA, UGA or UAG.

The correct answer is: Methionine

Question **63**

In which of the following types of inheritance, 25% of children inherit the disease even when

Not answered

Marked out of 1.00

Flag question

both parents are apparently healthy but share a similar genotype?

Select one:

- X linked recessive
- Mitochondrial
- Autosomal dominant
- Autosomal recessive
- X linked dominant

Autosomal recessive disorders are only manifest in the homozygous state and normally only affect individuals in one generation, usually in one sibship in a family. They affect both males and females equally. Offspring of parents who are heterozygous for the same autosomal recessive allele have a 1 in 4 chance of being homozygous for that allele. The less common an autosomal recessive allele, the greater the likelihood that the parents of a homozygote are consanguineous.

The correct answer is: Autosomal recessive

Question **64**

Not answered

Marked out of 1.00

Flag question

Which of the following is a clinical feature seen in Turner's syndrome (45X0)?

Select one:

- Webbed neck
- Narrow chest
- Prognathism
- Receding hairline
- Tall stature

Webbed neck with prominent low hairline, retrognathism, short stature and broad chest are features of Turner's.

The correct answer is: Webbed neck

Question **65**

Not answered

Marked out of 1.00

Flag question

In which of the following phases of the cell cycle does a neuron remain for the longest period of its life?

Select one:

- G1 phase
- G0 phase
- G2 phase
- S phase
- M phase

While certain cells pause or freeze the cycle temporarily and stay in G0, e.g. liver cells, neurons remain in G0 indefinitely. Collins, K et al. The cell cycle and cancer. Proceedings of the National Academy of Sciences 94: 2776-2778.

The correct answer is: G0 phase

Question **66**

Not answered

Marked out of 1.00

Flag question

A newly identified psychiatric disorder called 'somatomania' is found to be inherited in a simple Mendelian autosomal recessive fashion. If two patients with somatomania meet at a psychiatric ward, marry and give birth to two children, what are the chances (in percentage) that each child has inherited somatomania?

Select one:

- 33
- 25
- 100
- 75
- 50

In autosomal recessive disorders, children born to parents who are both diseased (not just carriers but diseased and so has homozygous recessive alleles) have 100% risk of inheritance.

The correct answer is: 100

Question **67**

Not answered

Marked out of 1.00

Flag question

Which of the following terms refers to the phenomenon by which one genetic alteration influences another genetic alteration eventually mitigating the effects of each other?

Select one:

- Imprinting
- Transition
- Crossing over
- Epistasis
- Synapsis

Gene-gene interaction between different alleles of different genes is called epistasis. This can occur at the same step or different stages of the same biochemical pathway. At times, the effect of one change may mitigate or modify the other.

The correct answer is: Epistasis

Question **68**

Not answered

Marked out of

In the human DNA, codons code for a specific

Select one:

1.00

Flag question

- Sugar
- Protein
- Phosphate
- Nuclei acid
- Amino acid

The genetic code is the relation between the sequence of bases in DNA (or its RNA transcripts) and the sequence of amino acids in proteins. Three nucleotides (codon) encode an amino acid. Proteins are built from a basic set of 20 amino acids, coded by several combinations of the four bases (Retrieved from http://www.bioscience-explained.org/ENvol4_1/pdf/spliceeng.pdf).

The correct answer is: Amino acid

Question **69**

Not answered

Marked out of 1.00

Flag question

The origin of single X chromosome in Turner's syndrome is most likely to be of

Select one:

- Dizygotic twin origin
- None of the listed
- Paternal origin
- Chromosome escaping random inactivation
- Maternal origin

As nearly 80% of meiotic error in Turner's is paternal, the remaining single X chromosome is usually of maternal origin.

The correct answer is: Maternal origin

Question **70**

Not answered

Marked out of 1.00

Flag question

The number of autosomes in a human somatic cell is

Select one:

- 2
- 22
- 44
- 23
- 46

Explanation:

An autosome is any chromosome other than the X and Y sex chromosomes. There are 44 autosomes in a human somatic cell.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: 44

Question **71**

Not answered

Marked out of 1.00

Flag question

For any particular trait, the pair of alleles of each parent separate and only one allele passes on from each parent to an offspring. Which of the following Mendelian principles refer to this phenomenon?

Select one:

- Law of common fate
- Law of uniformity
- Law of independent assortment
- Law of segregation
- Law of closure

Explanation:

The law of segregation (of alleles), the first of Mendel's laws, states that every somatic cell carries a pair of hereditary units ('alleles') for each character, and that at meiosis the pairs separate so that each gamete carries only unit from each pair.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Law of segregation

Question **72**

Not answered

Marked out of 1.00

Flag question

A patient with schizophrenia is more likely to marry someone with a psychiatric problem or substance use. This is termed as

Select one:

- Consanguineous mating
- Natural selection
- Genetic drift
- Pleiotropy
- Assortative mating

Explanation:

Assortative mating refers to a preference for sexual partners with similar phenotypes, as when humans select mates who tend to resemble themselves in wealth, social class, intelligence, or attractiveness.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Assortative mating

Question **73**

Not answered

Which of the following phenomenon has the highest heritability?

Marked out of 1.00

Flag question

Select one:

- IQ
- Neuroticism
- Extraversion
- Bipolar disorder
- Novelty seeking

Bipolar disorder has higher heritability than other listed traits.

The correct answer is: Bipolar disorder

Question **74**

Not answered

Marked out of 1.00

Flag question

In an attempt to find out if trichotillomania has a genetic basis, family history is collected by interviewing all possible first-degree relatives of 46 consecutive patients attending a hair disorder clinic. What type of study is this?

Select one:

- Family study
- Gene mapping
- Association study
- Linkage analysis
- Sib pair approach

Explanation:

In genetics, a family study (or kinship study) involves a comparison of the correlations between relatives of different known degrees of genetic relatedness on a measurable trait to estimate the heritability of that trait, the assumption being that the higher the heritability of the trait the more closely will the correlations correspond to degrees of relatedness. For example, because children share on average half their genes with each parent and with each sibling, the highest correlation between such relatives that would be expected for a trait with very high heritability is 0.5, and a substantially lower correlation would suggest low heritability.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Family study

Question **75**

Not answered

Marked out of 1.00

Flag question

Which of the following inheritance patterns includes maternal transmission but no paternal transmission and affects both sexes?

Select one:

- Mitochondrial
- Autosomal dominant
- X-linked dominant
- Genomic imprinting
-

X-linked recessive

Explanation:

Mitochondrial DNA is a form of DNA found in mitochondria, having its own genetic code, slightly different from the universal code of nuclear DNA. It is inherited only from mothers (because only the mother's ovum contributes cytoplasm to the zygote), it consists of several loops (each containing 37 genes), and it is present in every mitochondrion. Mutations in mitochondrial DNA have been linked to ageing and to several chronic degenerative diseases, including dystonia, Alzheimer disease, and diabetes mellitus.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Mitochondrial

Question **76**

Not answered

Marked out of 1.00

Flag question

Which of the following genes is NOT implicated in schizophrenia?

Select one:

- D aminoacid oxidase
- Dysbindin
- Monoamine oxidase
- Neuregulin
- COMT

MAO gene is not implicated; it is implicated in neuroticism and in anxiety disorders.

The correct answer is: Monoamine oxidase

Question **77**

Not answered

Marked out of 1.00

Flag question

Equatorial alignment of chromosomes is seen in which of the following phases of cell division?

Select one:

- Interphase
- Anaphase
- Telophase
- Prophase
- Metaphase

In metaphase, the chromosomes are aligned at the metaphase plate (a plane that is equally distant from the two spindle poles) - equatorial alignment.

The correct answer is: Metaphase

Question **78**

Not answered

Marked out of 1.00

Flag question

In which of the following mutations does the mutant codon specify a stop signal?

Select one:

- Missense mutation
- Silent mutation
- Nonsense mutation
- Inframe mutation
- Frameshift mutation

A silent mutation causes no change in protein product - this is possible because a single amino acid is often coded by more than one triplet sequence. In a silent mutation, one triplet sequence is replaced by a different sequence but without changing amino acid product. In missense mutation, the new mutant codon specifies a different amino acid with variable effects on final protein product. For example, haemophilia, sickle cell anaemia. In non sense mutation, the new codon is UUA UGA or UAG, which signals 'stop' to the amino acid sequence resulting in a nonfunctional protein.

The correct answer is: Nonsense mutation

Question **79**

Not answered

Marked out of 1.00

Flag question

CADASIL is associated with a mutation in which of the following genes?

Select one:

- Dysbindin
- LRRK2
- Parkin gene
- Notch3
- COMT

An unusual form of subcortical small vessel disease occurs in the rare inherited condition of CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy) in which a mutation in the Notch3 gene results in deposition of granular material in the walls of small arteries and arterioles, narrowing of vessel lumens and destruction of smooth muscle cells in their walls. (Ref: Vascular dementia: a pragmatic review. Adv Psychiatric Treatment Sept 2012; 372-380)

The correct answer is: Notch3

Question **80**

Not answered

Marked out of 1.00

Flag question

Which type of genetic study uses cases and controls or families with internal controls?

Select one:

- Segregation studies
- Family studies
- Association studies

- Linkage analysis
- Twin studies

The simplest scenario of genetic association analysis is to study the genotype frequency of a polymorphism in a group of independent patients (cases) and a group of normal individuals (controls). They might be designed as simple case-control studies, or even better if they make use of family-based internal controls.

The correct answer is: Association studies

Question **81**

Not answered

Marked out of 1.00

Flag question

Which of the following gives the value of one centimorgan?

Select one:

- 100 base pairs of DNA
- 1000 basepairs of DNA
- 100% recombination frequency between two loci
- 1% recombination frequency between two loci
- 10 base pairs of DNA

Explanation:

The evidence for linkage between two loci depends on the recombination frequency between them. Recombination frequency is measured by the recombination fraction, equal to the genetic distance between the two loci. 1% recombination equals 1 centimorgan (cM) in the genetic distance and, on average, covers a physical distance of about 1 megabase of DNA.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 988.

The correct answer is: 1% recombination frequency between two loci

Question **82**

Not answered

Marked out of 1.00

Flag question

In premutant carriers of fragile X expansion, which of the following features may be seen?

Select one:

- psychotic episodes
- increased risk for intention tremor and ataxia
- no clinical features
- mental retardation
- macroorchidism

Men who are not retarded but carry an increased number of CGG repeats in the FMR1 locus are called premutation carriers. Though premutant carriers were long thought to be free from clinical features, it is now known that they are at increased risk for developing intention tremor and ataxia especially after middle age. Likewise, women who are premutation carriers

(55-200 CGG repeats) are at increased risk of premature ovarian failure and/or mild cognitive or behavioral abnormalities.

The correct answer is: increased risk for intention tremor and ataxia

Question **83**

Not answered

Marked out of 1.00

Flag question

In which of the following conditions chromosomal duplication is seen?

Select one:

- Cri-du-chat Syndrome
- Foetal Alcohol Syndrome
- Turner's Syndrome
- Reye's Syndrome
- Trisomy 21

Reye's and FAS are not chromosomal disorders. Cri-du-chat is a deletion syndrome while Turner's is a monosomy with non-disjunction (no duplication).

The correct answer is: Trisomy 21

Question **84**

Not answered

Marked out of 1.00

Flag question

The gene polymorphism involved in serotonin transporter variations is

Select one:

- 5q31
- SLC6A4
- XLMR
- Xq 26-27
- HGPRT

Explanation:

The 5-hydroxytryptamine transporter (5HTT), encoded by SLC6A4 (solute carrier family 6, member 4), also known as sodium-dependent serotonin transporter, is found in the plasmatic membrane and transports the neurotransmitter serotonin from the synaptic cleft into presynaptic neurons for recycling, thus terminating its action. 5HTTLPR-VNTR is one of the described genetic variants of the SLC6A4 gene. Located on chromosome 17q11.2, this polymorphism gives rise to two different alleles: a long allele (Long, L) and a short allele (Short, S). Population genetic studies have shown that the S allele presence decreases the 5HTT promoter gene's transcriptional efficiency, resulting in a lower serotonin transporter binding and its uptake. Accordingly, this genetic alteration may increase the risk of susceptibility to psychiatric disorders, such as major depression disorder.

Ref: Fratelli C et al. 5HTTLPR Genetic Variant and Major Depressive Disorder: A Review. Genes. 2020 Nov;11(11):1260.

The correct answer is: SLC6A4

Question **85**

Not answered

Marked out of 1.00

Flag question

A 77-year-old Vietnam War veteran presents with shooting pain in his back. On neurological examination, you find light near dissociation of Argyll Robertson Pupil. Which of the following is true?

Select one:

- Consensual light reflex will be present
- Both light and accommodation reflex will be present
- Light reflex will be present
- Accommodation reflex will be present
- Afferent pupillary defect will be seen

ARP - Argyll Robertson Pupil (note: Accommodation Reflex Present -light reflex absent) is due to an afferent defect in the pupillary reflex pathway - most probably a pretectal lesion.

The correct answer is: Accommodation reflex will be present

Question **86**

Not answered

Marked out of 1.00

Flag question

If both parents are carriers of an autosomal recessive illness, then the probability of inheritance in children is

Select one:

- 1 in 4
- 1 in 2
- 100%
- 1 in 3
- 0%

If both parents carry ONE autosomal recessive gene each (both are carriers), there is a 1 in 4 chance of the child developing the disease. If both parents carry TWO autosomal recessive genes each (both are diseased), there is a 100% chance of the child developing the disease.

The correct answer is: 1 in 4

Question **87**

Not answered

Marked out of 1.00

Flag question

Which of the following best describes the term genotype?

Select one:

- The genetic makeup of a family with a specific disorder
- An unobservable marker of familial risk for a condition
- The detectable characteristics of an individual organism
- A numerical representation of all of the potential combinations of parental traits.
- The sum total of all the genes that is inherited by an organism.

The term genotype describes the actual set of genes carried by an organism. In 1909, the clarifying distinction between the concept of "genotype" and the "phenotype" was provided by the Danish botanist Wilhelm Johanssen. Gottesman and Shields described "endophenotypes" as internal phenotypes discoverable by a "biochemical test or microscopic examination."

The correct answer is: The sum total of all the genes that is inherited by an organism.

Question **88**

Not answered

Marked out of 1.00

Flag question

A mutation in the shank3 gene is associated with which of the following features?

Select one:

- Parkinsonism
- Autism
- Bulimia
- Alcohol dependence
- Anorexia nervosa

SHANK3 or 'SH3 and multiple ankyrin repeat domains 3' is located at 22q13.3. 22q13.3 deletion syndrome (called Phelan-McDermid syndrome) reduces cell-to-cell communication between neurons caused by a lack of SHANK3 protein contributing to the developmental delay, intellectual disability, and absent or severely delayed speech. At least seven SHANK3 gene mutations have been found in people who have the features of autism or similar conditions known as autism spectrum disorders (Retrieved from http://ghr.nlm.nih.gov/gene/SHANK3).

The correct answer is: Autism

Question **89**

Not answered

Marked out of 1.00

Flag question

A 24-year-old woman has a 32-year-old brother with schizophrenia. What is the risk (expressed as percentage) of her developing schizophrenia according to DSM criteria?

Select one:

- 13
- 6
- 10
- 2
- 16

The lifetime risk of schizophrenia in the sibling of a patient is 9-10%.

The correct answer is: 10

Question **90**

Not answered

Lucy is Lenny's half sister. The proportion of genes that might be shared between Lucy and Lenny is

Marked out of 1.00

Flag question

Select one:

- 50
- 33
- 25
- 20
- 12

First-degree relatives share one-half of their genes. Second-degree relatives share one-fourth, and third-degree relatives share one-eighth.

The correct answer is: 25

Question **91**

Not answered

Marked out of 1.00

Flag question

Which of the following disorders is inherited via X-linked dominant inheritance?

Select one:

- Cerebellar vermis herniation with hydrocephalus.
- Brain tumours, epilepsy, intellectual disability and multiple renal cysts.
- A 3 yr old girl with intellectual disability and stereotyped hand movements around the midline of her body.
- A 2 yr old boy with repeated respiratory infections, puppet-like face, hypothyroid profile and poor cognitive development born to a mother of advanced maternal age.
- Pseudohypertrophy of calf muscles with normal IQ.

Explanation:

Rett syndrome is believed to be caused bya **dominant X-linked gene.** It is a degenerativedisorder that almost exclusively affects **females.** Between ages 1- 4 years, purposeful hand movements and spoken language are lost, and stereotypical hand movements (hand-wringing) and autistic signs are seen. Other features include ataxia, facial grimacing, teeth-grinding, and loss of speech.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 103.

The correct answer is: A 3 yr old girl with intellectual disability and stereotyped hand movements around the midline of her body.

Question **92**

Not answered

Marked out of 1.00

Flag question

Which of the following is least likely regarding diseases that show the phenomenon of anticipation?

Select one:

- Dominant mode of transmission is noted
- The age of onset and severity of illness reduces with successive generations
- Trinucleotide expansion is seen

- Infants are normal at birth
- Most of them are neuro-degenerative disorders

With successive generations, if the onset is earlier and the outcome becomes more severe in a genetic disease, the term anticipation is employed to describe the inheritance.

The correct answer is: The age of onset and severity of illness reduces with successive generations

Question **93**

Not answered

Marked out of 1.00

Flag question

Which of the following genetic variants of Down syndrome shows less cognitive impairment and milder features than other variants?

Select one:

- Mosaicism
- Full trisomy of chromosome 21
- Duplication of a portion of chromosome 21
- Both translocation and mosaicism
- Robertsonian translocation

Explanation:

Mosaicism (a mixture of normal and trisomic cell lines) occurs in approximately 2-5% of individuals with Down syndrome (DS). IQ can be the 70s (IQ in adults with DS is most often below 50), and the physical abnormalities may be less marked.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 806.

Antonarakis SE et al. Down syndrome. Nature Reviews Disease Primers. 2020 Feb 6;6(1):1-20.

The correct answer is: Mosaicism

Question **94**

Not answered

Marked out of 1.00

Flag question

Serotonin transporter promoter region shows short and long allele variants as a polymorphism. Which of the following statements about polymorphisms, in general, is true?

Select one:

- Generally occur at a rate above 6% in population
- Maintained by evolutionary processes
- Produce a functional defect in the subject
- Cannot be used for linkage analysis
- Can be seen only in exons

Polymorphism refers to variations in genetic make-up at a particular locus noted in general, apparently healthy population. To be defined as polymorphism, the variant must occur in at

least 1% of the total population and must be associated with normal (not abnormal i.e. disease-causing) expression of the final phenotype. This excludes spontaneous mutations that are random and so cannot simultaneously occur in such significant (1%) proportion of total population. ABO blood groups are good examples of polymorphism expressed in protein products of genes. It is assumed that evolutionary selection serves to maintain polymorphisms.

The correct answer is: Maintained by evolutionary processes

Question **95**

Not answered

Marked out of 1.00

Flag question

In Robertsonian translocation, the fusion usually occurs at the

Select one:

- Cytoplasm
- Centromere
- Non sense mutations
- Stop codons
- Aneuploidy

A Robertsonian translocation is a type of nonreciprocal translocation involving two homologous (paired chromosomes) or non-homologous chromosomes (i.e.,. two different chromosomes, not belonging to a homologous pair). A feature of chromosomes that are commonly found to undergo such translocations is that they possess an acrocentric centromere (having a large arm with the majority of genes, and a short arm with sparse number of genes). During a Robertsonian translocation, chromosomes break at their centromeres and the two long arms fuse to form a metacentric chromosome. The short arms also join to form a smaller chromosome, but this small piece gets lost over a few cell divisions.

The correct answer is: Centromere

Question **96**

Not answered

Marked out of 1.00

Flag question

In a patient with macroorchidism, large ears, prominent jaw, a high-pitched voice, and mental retardation, which of the following trinucleotide repeats could be seen?

Select one:

- CCT
- CTG
- CGG
- CTT
- CAG

The above description suits Fragile X syndrome. This X-linked condition accounts for more cases of mental retardation in males than any other condition except Down syndrome with a frequency of 1 in 4000. It can affect females but 50% less frequently than in males. A fragile site near the tip of the long arm of the X chromosome was initially suspected. Now it is known that fragile X results from an expansion of a trinucleotide repeat (CGG) near a gene called

FMR1.

The correct answer is: CGG

Question **97**

Not answered

Marked out of 1.00

Flag question

Right homonymous hemianopia is caused by which of the following lesions?

Select one:

- Optic chiasma lesion
- Lesions of right optic radiation
- Lesions of left optic radiation
- Lesions of right optic nerve
- Lesions of left optic nerve

Lesions of the left-sided optic tract, lateral geniculate body, optic radiations, and striate cortex (any retro chiasmatic structure) can cause right homonymous hemianopia.

The correct answer is: Lesions of left optic radiation

Question **98**

Not answered

Marked out of 1.00

Flag question

What percentage of the human genome is considered to be active with coding sequences?

Select one:

- 20
- 2
- 30
- 40
- 10

The proportion of coding and noncoding DNA within genomes varies widely among organisms. More than 98% of the human genome does not encode protein sequences, including most sequences within introns and most intergenic DNA. Nature Reviews Genetics 11, 559-571 (August 2010)

The correct answer is: 2

Question **99**

Not answered

Marked out of 1.00

Flag question

The most common feature of Prader-Willi syndrome is

Select one:

- Epilepsy
- Repeated hand wringing
- Large testicles
- Short stature

- Simian palmar crease

The features of Prader-Willi syndrome includes obesity, small hands and feet, microorchidism, cryptorchidism (hypogonadism), short stature, almond-shaped eyes, flat face, prominent forehead and bitemporal narrowing and orthopaedic problems such as congenital dislocation of hip, scoliosis etc are seen as the child develops.

The correct answer is: Short stature

Question **100**

Not answered

Marked out of 1.00

Flag question

In which of the following phases of the cell cycle does cell division occur?

Select one:

- G1 phase
- G0 phase
- G2 phase
- S phase
- M phase

Each cell undergoes a natural cycle in terms of its replication and nucleic acid synthetic activity. The cell cycle consists of four separate phases: G1 phase, S phase, G2 phase and M phase. G1 stands for growth phase 1, S for synthetic phase, G2 for growth phase 2 and M phase for mitosis phase. Cells in the quiescent G0 phase of the cycle are stimulated by the growth factors (e.g. EGF, epithelial growth factor; PDGF, platelet-derived growth factor; IGF, insulin-like growth factor) and result in activation of transcription factors and lead to the initiation of DNA synthesis, followed by mitosis and cell division. Thus from G0 the cell moves on to G1 when the chromosomes are prepared for replication. This is followed by the synthetic (S) phase, when the 46 chromosomes are duplicated into chromatids, followed by another gap phase (G2), which eventually leads to mitosis (M). Collins, K et al. The cell cycle and cancer. Proceedings of the National Academy of Sciences 94: 2776-2778.

The correct answer is: M phase

Question **101**

Not answered

Marked out of 1.00

Flag question

Which of the following criteria must be satisfied for a trait to be called an endophenotype?

Select one:

- Must be independent of the clinical state
- Must be associated with a candidate gene
- All of the listed options
- Must be co segregating with actual illness
- Must have biological plausibility

To be termed as an endophenotype, Gottesman suggested certain criteria to be satisfied by an identified disease marker. These are as follows: Must be associated with a candidate gene or region Must be present with a high relative risk in relatives, thus co segregating with

actual illness Must be a parameter associated with disease with biological plausibility Must be independent of the clinical state (i.e. must not be a state but a trait marker) Must be heritable Must be present in relatives more often than general population

The correct answer is: All of the listed options

Question **102**

Not answered

Marked out of 1.00

Flag question

Soon after gamete fusion and fertilisation, nondisjunction of chromosomes takes place in an early mitotic cell division in chromosome 21. Which of the following chromosomal makeup will be seen in the foetus?

Select one:

- Triploidy
- 21 monosomy
- Euploidy
- 21 trisomy mosaic
- 21 trisomy

Explanation:

Nondisjunction occurring after fertilisation in any cell division results in mosaicism, a condition in which there are both normal and trisomic cells in various tissues. Mosaicism occurs in approximately 2-5% of individuals with Down syndrome (DS).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 101.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 806.

The correct answer is: 21 trisomy mosaic

Question **103**

Not answered

Marked out of 1.00

Flag question

A common method used to measure recombination fraction in a linkage study is

Select one:

- Delta score
- Log of odds score
- Predicted recombination frequency
- Genetic distance
- Observed recombination frequency

LOD or logarithm of odds score is a measure of the statistical strength of association used in a genetic linkage analysis. The value of the LOD score is the ratio between the probability of obtaining a result (of association) if the two loci are linked to the probability of obtaining the results if the two loci are not linked. Higher the score, more the likelihood of a true association.

The correct answer is: Log of odds score

Question **104**

Not answered

Marked out of 1.00

Flag question

Deletion in the paternally derived chromosome 15q11-q13 leads to which of the following disorders?

Select one:

- Tuberous sclerosis
- Huntington disease
- Di George syndrome
- Angelman syndrome
- Prader-Willi syndrome

Explanation:

Prader-Willi syndrome is a rare genetic disorder associated with **microdeletion on chromosome 15** (q11-13), usually on the **paternal** chromosome. It is the complement of Angelman syndrome.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 808.

The correct answer is: Prader-Willi syndrome

Question **105**

Not answered

Marked out of 1.00

Flag question

In a family with the X-linked recessive transmission of a genetic disease, there are two daughters and one son born to a diseased father and normal non-carrier mother. One of the two daughters exhibits a moderate degree of the disorder while the other is a silent carrier. Which of the following can explain the above?

Select one:

- The daughter is biologically male
- Combined X linked dominant and recessive transmission
- Diseased daughter is the unaffected carrier
- The disease cannot be X linked recessive
- Unfavourable lyonisation of X chromosomes

If a recessive disease-causing mutation occurs on the single X chromosome of a man, this is sufficient to cause disease as another X chromosome is not available to compensate any deficiencies. As females have two copies of the X chromosome, they need a double identical mutation for disease expression which is extremely rare. But during random X inactivation if most X chromosomes carrying normal alleles are inactivated (called unfavourable Lyonisation), then these females can manifest the disease phenotype " termed as manifesting heterozygotes. But nevertheless the severity of expressed disease is mild and can go unnoticed too.

The correct answer is: Unfavourable lyonisation of X chromosomes

Question **106**

Not answered

Marked out of 1.00

Flag question

The commonest cardiac malformation seen in Down syndrome is

Select one:

- Endocardial cushion defect
- Pulmonary stenosis
- Tetralogy of Fallot
- Atrial ostium secundum defect
- Atrial stenosis

Congenital heart defects are common in Down (40-50%); they are frequently observed in patients with Down syndrome who are hospitalized (62%) and are a common cause of death in this aneuploidy in the first 2 years of life. The most common congenital heart defects are endocardial cushion defect (43%), ventricular septal defect (32%), secundum atrial septal defect (10%), tetralogy of Fallot (6%), and isolated patent ductus arteriosus (4%). PDA is seen in 16% while pulmonary stenosis in 9% (http://emedicine.medscape.com/article/943216-overview).

The correct answer is: Endocardial cushion defect

Question **107**

Not answered

Marked out of 1.00

Flag question

Which of the following best describes the commonest type of mutations seen in thalassemia?

Select one:

- Nonsense
- Inframe
- Missense
- Frame shift
- Silent

Beta thalassemia is the commonest single gene disorder in the world. It is generally considered as an autosomal recessive disorder as the inheritance of two mutant ß-globin genes is necessary to produce a notable clinical illness. It is characterized by mostly missense or somewhat less frequently nonsense mutations in the [beta]-globin gene (HBB), which reduce or completely abolish gene expression. Heterozygous inheritance of a [beta]-thalassemia mutation results in a trait condition, sometimes termed [beta]-thalassemia minor. Homozygous inheritance results in [beta]-thalassemia major (absent Hb A production) or [beta]-thalassemia intermedia (severely reduced Hb A production), which are associated with moderate to severe hemolytic anemia.

The correct answer is: Missense

Question **108**

Not answered

Marked out of 1.00

A 45-year-old man is diagnosed with cognitive impairment. The family history suggests that individuals in the most recent generations of the family pedigree have developed a myriad of psychiatric disturbances at an earlier age and with greater severity than previous generations. Choose one correct diagnosis.

Flag question

Select one:

- Fragile X syndrome
- Huntington syndrome
- Motor neuron disease.
- Velocardiofacial syndrome
- Frontotemporal dementia

Huntington's disease is inherited in an autosomal dominant manner with full penetrance and a prevalence rate of about 5 per 100,000. The gene responsible is an expanded and unstable CAG trinucleotide repeat on the short arm of chromosome 4 - 4p16.3. This results in translation of an extended glutamine sequence in huntingtin, the protein product of the gene. Huntingtin is expressed throughout the body. Its function is unclear.Though slightly unusual for a genetic disease; the onset is usually between 30 and 50 years of age. Most adult-onset HD cases have CAG expansions of 40-55 repeats, while greater expansions (and more than 70 repeats) are seen in childhood-onset HD. The phenomenon of anticipation, as described in the above scenario, is well demonstrated in Huntington's.

The correct answer is: Huntington syndrome

Question **109**

Not answered

Marked out of 1.00

Flag question

Which of the following is true concerning a woman who has an autosomal recessive disorder but her husband is genotypically normal?

Select one:

- Only her daughters will receive the mutant gene
- All her children will have the disease phenotype
- None of her children will have the disease phenotype
- All her daughters will be carriers, all her sons will have the disease
- All her sons will be carriers, all her daughters will have the disease

As all of the children will be carriers, none will have the expressed phenotype in normal circumstances.

The correct answer is: None of her children will have the disease phenotype

Question **110**

Not answered

Marked out of 1.00

Flag question

In a family, father suffers from an autosomal dominant disease. Mother is a homozygous normal individual. Which of the following correctly describes the risk of this couple having a child with the autosomal dominant disorder?

Select one:

- It depends on the age of the father at time of birth
- It depends on the birthorder
- It depends on the sex of the child
- The risk is 50% irrespective of sex

- The risk increases with incomplete penetrance

Each cell contains two copies of all the autosomes. An autosomal dominant disorder occurs when one of the two copies has a mutation and the protein produced by the normal form of the gene cannot compensate. So the mutant allele becomes dominant over the normal allele and results in disease expression. In this case, a heterozygous individual who has two different forms (or alleles) of the same gene will manifest the disease. The offspring of heterozygotes have a 50% chance of inheriting the chromosome carrying the disease allele, and therefore also of having the disease. If both parents are heterozygous, the recurrence risk is 75%. This risk reduces with incomplete penetrance but is not affected by birth order or sex.

The correct answer is: The risk is 50% irrespective of sex

Question **111**

Not answered

Marked out of 1.00

Flag question

If a disease has 75% heritability, then which of the following statement is correct?

Select one:

- 75% of the diseased patients have inherited the condition
- 75% of the disease related genes are passed on from one generation to another
- The risk of the disease in first degree relatives of patients is 75%
- 75% of the features seen in a diseased patient is due to genetic component
- 75% of the liability for the disease is inherited

Heritability is defined by the proportion of variation of an observed feature that is due to genetic factors.

The correct answer is: 75% of the liability for the disease is inherited

Question **112**

Not answered

Marked out of 1.00

Flag question

A man suffers from an autosomal recessive disorder and has mild learning difficulties and special dietary requirements. Which of the following enzymes is affected as a result of the mutated gene?

Select one:

- Phenylalanine hydroxylase
- Phenylalanine decarboxylase
- Alcohol dehydrogenase
- Hypoxanthine guanine phosphoribosyl hydroxylase
- Hypoxanthine guanine phosphoribosyl transferase

The diagnosis is phenylketonuria, which is an inborn metabolic disorder with autosomal recessive inheritance. It is a complex metabolic condition caused by mutations of the phenylalanine hydroxylase gene.

The correct answer is: Phenylalanine hydroxylase

Question **113**

Not answered

Marked out of 1.00

Flag question

The mechanism of meiosis fits with which of the following descriptions?

Select one:

- Occurs in all tissues
- Has a short prophase
- Produces daughter cells identical to parent cells
- Produces diploid somatic cells
- Includes recombination

During the early stages of cell division in meiosis, two chromosomes of a homologous pair may exchange segments - this is called recombination.

The correct answer is: Includes recombination

Finish review

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